Linear Algebra: Eigenvalues and Eigenvectors

Lecture Summary

1 Introduction and Definitions

A linear transformation represented by an $n \times n$ matrix A typically rotates and scales vectors. However, certain ”special” vectors remain on their own span after the transformation.

1.1 The Eigenvalue Equation

An **eigenvector** of an $n \times n$ matrix A is a non-zero vector x such that when A is applied to x, the result is a scalar multiple of x. The formal definition is given by:

$$Ax = \lambda x \tag{1}$$

Where:

- A is an $n \times n$ square matrix.
- x is the **eigenvector** (must be non-zero, $x \neq 0$).
- λ is the **eigenvalue** corresponding to x.

2 Finding Eigenvalues and Eigenvectors

2.1 The Characteristic Equation

To find the eigenvalues λ, we rewrite the equation as $(A - \lambda I)x = 0$. For a non-trivial solution ($x \neq 0$) to exist, the matrix $(A - \lambda I)$ must be singular. Thus, we solve the **characteristic equation**:

$$\det(A - \lambda I) = 0 \tag{2}$$

The determinant $|A - \lambda I|$ is a polynomial in λ called the **characteristic polynomial**. The roots of this polynomial are the eigenvalues of A.

2.2 Finding Eigenvectors

Once the eigenvalues λ_i are found, the corresponding eigenvectors $x^{(i)}$ are found by solving the homogeneous system:

$$(A - \lambda_i I)x^{(i)} = 0 \tag{3}$$

This is typically done using Gaussian elimination or by setting free variables to determine a basis for the eigenspace.

3 Properties of Eigenvalues

3.1 Trace of a Matrix

The **trace** of a square matrix A, denoted tr(A), is the sum of the elements on its main diagonal. A fundamental property is that the trace is equal to the sum of the eigenvalues:

$$\text{tr}(A) = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i \tag{4}$$

3.2 General Observations

- An $n \times n$ matrix always has n eigenvalues (counting multiplicity).
- Eigenvalues can be real, distinct, repeated, or complex conjugates.
- Eigenvectors are not unique; any scalar multiple of an eigenvector is also an eigenvector.

4 Diagonalization (Eigendecomposition)

Diagonalization is the process of decomposing a matrix A into a product of matrices involving its eigenvalues and eigenvectors.

4.1 Decomposition Formula

If A has n linearly independent eigenvectors, it can be diagonalized as:

$$A = PDP^{-1} \tag{5}$$

Where:

- P is the **modal matrix** whose columns are the eigenvectors of A.
- D is the **spectral matrix**, a diagonal matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ on the main diagonal.

5 Matrix Functions

One of the most powerful applications of diagonalization is the ease of computing matrix powers and functions.

5.1 Matrix Powers

If $A = PDP^{-1}$, then:

$$A^n = PD^nP^{-1} \tag{6}$$

Since D is diagonal, D^n is simply the diagonal matrix where each entry λ_i is raised to the power n.

5.2 Elementary Functions

Using Taylor series expansions, we can define functions like the matrix exponential e^A:

$$e^A = Pe^D P^{-1} = P \begin{pmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{pmatrix} P^{-1} \tag{7}$$

Similarly, this applies to functions like $\ln(A)$, $\sin(A)$, etc., by applying the function to the diagonal entries of D.

6 Applications

Eigen-analysis is fundamental in various fields, including:

- **Differential Equations:** Solving systems of linear differential equations.
- **PageRank:** Used by search engines to rank web pages.
- **Eigenfaces:** Used in computer vision and facial recognition (Principal Component Analysis).